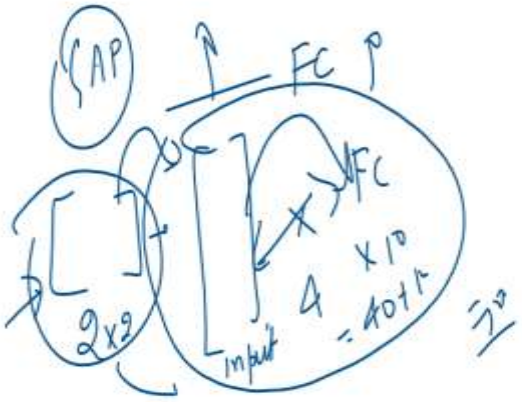
Optimizers



Optimizers

Week 2

loss function → min

$\mathcal{L}(W, b)$

Gradient Descent "optimization"

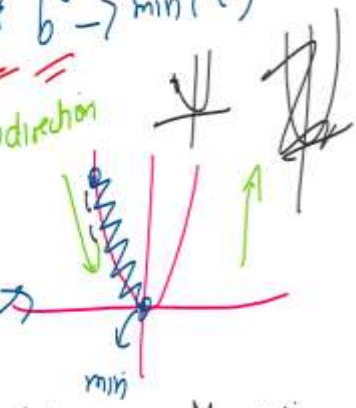
$W^* \quad b^* \rightarrow \min(\mathcal{L})$

Gradient Descent

$W_t = W_{t-1} - \eta \nabla L_{W_t}$

opt → η

direction → ∇L_{W_t}



W^4

learning rate : step size → Magnitude

0.1, 0.001, 0.0001

10

t+1, t

Momentum Based Gradient Descent

0.1, 0.001, 0.0001

increasing speed

$W_1 = W_0 - \eta \nabla W_0$

$W_{t+1} = W_{t-1} - \eta U_{t-1}$ → history of gradients

Weight

$U_t = \beta U_{t-1} + \nabla W_t$

Gradient

$U_5 = U_0 - U_1 - U_2 - U_3$

$U_0 = \nabla W_0$

0.9×0.9

$U_1 = \beta U_0 + \nabla W_1$

$U_2 = \beta U_1 + \nabla W_2$

$\beta = 0.9, 0.99, 0.009$

$1 + 1 + 1 + 1$

$1 + 0.9 + 0.09 + 0.98$

GD	MGD
$W_1 = W_0 - \eta \nabla W_0$	$W_1 = W_0 - \eta U_0$
$W_2 = W_1 - \eta \nabla W_1$	$U_0 = \beta U_{-1} + \nabla W_0$ ($U_{-1} = 0$)
∇L_{W_0}	$W_2 = W_1 - \eta U_1$
	$U_1 = \beta U_0 + \nabla W_1$
	β

MGD

$$W_1 = W_0 - \eta U_0$$

$$U_0 = \beta U_{-1} + \nabla L_{W_0}$$

$$= \nabla L_{W_0}$$

$$W_2 = W_1 - \eta U_1$$

$$U_1 = \beta U_0 + \nabla L_{W_1} = \beta \nabla L_{W_0} + \nabla L_{W_1}$$

$$W_3 = W_2 - \eta U_2$$

$$U_2 = \beta U_1 + \nabla L_{W_2} = \beta^2 \nabla L_{W_0} + \beta \nabla L_{W_1} + \nabla L_{W_2}$$

$(0.9)^2$ scaling factor

1+1

1+1+0.001

$$\beta^3 \nabla L_{W_0} + \beta^2 \nabla L_{W_1} + \beta \nabla L_{W_2} + \nabla L_{W_3}$$

→ try − η

$$W_t = W_{t-1} - \eta \nabla L_{W_{t-1}}$$

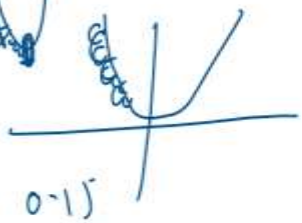
w_{t-1}

Adagrad

$$w_t = w_{t-1} - \frac{\eta}{\sqrt{V_t + \epsilon}} \nabla L_{w_{t-1}}$$

η is decreasing in every iteration

$$V_t = V_{t-1} + \left(\nabla L_{w_t}\right)^2$$



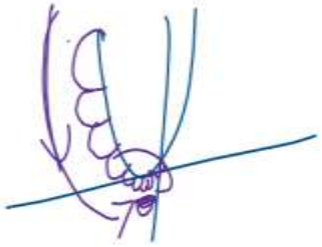
$\eta_1 = 1$

$\eta_2 = 0.5 \quad \eta_3 = 0.25 \quad \eta_4 = 0.15$

→ large η 0

Speed ↑

Acc ↑



Adagrad

$$w_t = w_{t-1} - \frac{\eta}{\sqrt{V_t + \epsilon}} \nabla L_{w_{t-1}}$$

$\frac{\eta}{0}$

very small no.

$$V_t = V_{t-1} + \left(\nabla L_{w_t}\right)^2$$

$\eta \downarrow$

learning rate is constantly decreasing - Prob

$$w_1 = w_0 - \frac{\eta}{\sqrt{V_0}} \nabla L_{w_0}$$

$$V_0 = V_{-1} + \left(\nabla L_{w_0}\right)^2 = \left(\nabla L_{w_0}\right)^2$$

$$V_1 = V_0 + \left(\nabla L_{w_1}\right)^2 = \left(\nabla L_{w_0}\right)^2 + \left(\nabla L_{w_1}\right)^2$$

$$V_2 = \left(\nabla L_{w_0}\right)^2 + \left(\nabla L_{w_1}\right)^2 + \left(\nabla L_{w_2}\right)^2 \Big\} \text{ History of Grad}$$

③ RMSProp

$$W_t = W_{t-1} - \frac{\eta \nabla_{W_t} L}{\sqrt{V_t + \varepsilon}}$$

$\eta \rightarrow$ still reduces

initial $\eta \rightarrow$ Hyperparameter

$$V_t = \beta V_{t-1} + (1-\beta)(\nabla_t W)^2$$

$\beta = 0.9$

$$V_0 = \beta V_{-1} + (1-\beta)(\nabla_{W_0} L)^2$$

$$(1-\beta)(\nabla_{W_0} L)^2$$

$$0.1(\nabla_{W_0} L)^2$$

V_t

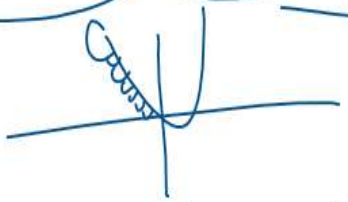
$$V_1 = \beta V_0 + (1-\beta)(\nabla_{W_1} L)^2$$

$$0.9(V_0) + (0.1)(\nabla_{W_1} L)^2$$

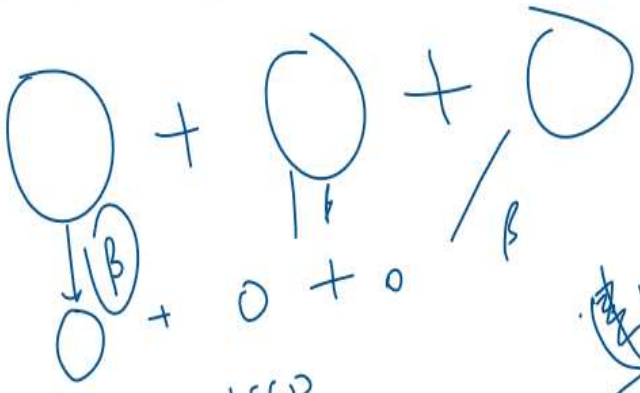
V_t

∴ RMS prop controls the pace

$\eta = 0.1$



logscale $\left[\begin{array}{l} \eta = 0.1, 0.01, \ldots \\ \beta = 0.9, 0.09, \ldots \end{array} \right.$



Adagrad + SGD

Adam → Combining RMS prop + Momentum

Adaptive Momentum

Opt

η → RMS

∇L → Momentum

→ Controlled Steps with faster convergence with accuracy

Adam $W_t = W_{t-1} - \eta \nabla L$

$$W_{t+1} = W_t - \frac{\eta}{\sqrt{V_t + \epsilon}} U_{t-1}$$

→ faster Speed + Accuracy

→ Controlled Step → Never Miss Min

$$V_t = \beta V_{t-1} + (1-\beta)(\nabla L_{W_t})^2$$ → RMS prop

$$U_t = \beta U_{t-1} + (1-\beta) \nabla L_{W_t}$$

$0 + 0$

$0 + 0$

Bias Correction bring β

0.9

0.7

V_t → $\hat{V}_t = \frac{V_t}{1-\beta}$

U_t → Momentum

$\hat{m}_t = \frac{m_t}{1-\beta}$ stable

$\beta = 0.9$

β → 0.7 - 0.09

0.7

W^0 → W^L